# BACKWARD ERROR OF THE DC ALGORITHM

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Abstract. We study the DC algorithm (DCA) in floating-point arithmetic. DCA minimizes $f = g - h$, a difference of convex functions, by solving at each step a convex subproblem built from the slope of $h$ at the current point. In the quadratic case this subproblem is the linear system $Qx = y_k$. The matrix $Q$ can be ill-conditioned up to $10^{16}$, so the computed solution can be far from the exact one in the usual forward sense — yet DCA converges in practice. This paper explains why.

Instead of asking how far the computed iterate is from the exact one, we ask: for which slightly perturbed problem is the computed iterate *exact*? When the perturbation is on the order of the rounding unit $u$, the computed answer is as good as any computer can produce, since the input data itself is rounded to order $u$. We prove a per-step backward-error bound of order $u$ and a composition bound $R_K \leq \frac{u}{2}(\max S_k)\,\mathrm{TV}_K$ over $K$ steps, with no alignment or feasibility assumption. The consequence is that the final iterate nearly satisfies the first-order optimality condition of the true problem. We state precisely what is proved, what is observed numerically, and what remains open; nothing is claimed beyond the evidence.

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## 1. The refined slogan, stated conditionally

**Remark 1.1** (Labels)**.** Every statement below carries one of the following labels, each with its place in the confidence scale: *[Theorem]* — proved by hand; the highest confidence. *[Computed]* — verified by code, not a proof. *[Standard]* — established literature. *[Observed]* — measured, with a proposed mechanism. *[Conjecture]* — open. *[Conditional]* — true only under a stated feasibility hypothesis. *[Erratum]* — corrects an earlier version. *[Unreproduced]* — numerical value whose scripts and data are not in the accompanying materials. *[Reproduced]* — numerical value re-implemented from the protocol text and re-run; scripts, data, and seed are in the accompanying materials (`reproduction/`). *[Solver-sensitive]* — numerical value that depends on solver details lost with the original scripts (the Newton stopping rule); the paper's value is pinned to the original run, and the reproduction packet's value is reported alongside. *[Proved — agent level]* — proved by an agent; pending author review. *[Proved — agent level; pending author review; Lean verification pending.]* — agent-level proof done; author review and Lean machine-check still pending. *[SKETCH]* — proof sketch,

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not a complete proof. *[Open]* — open problem or undetermined constant. *[Proposed at agent level; pending author review.]* — a proposed correction, not yet approved by the author. No proof is fabricated; review-supplied arguments are marked as such.

The purpose of this section is to pose Wilkinson's question for the DC algorithm, to define the gauge-perturbed pair that answers it, and to isolate the feasibility condition on which the answer depends. We begin by elaborating on the slogan sketched in the abstract.

Wilkinson's backward-error analysis asks: for what *perturbed* problem is the computed answer *exact*? The famous answer for linear systems is due to Wilkinson himself: Gaussian elimination with partial pivoting computes the exact solution of $(A+\Delta A)x = b+\Delta b$ with $\|\Delta A\| \lesssim u\|A\|$ [1] [Standard], where $u$ denotes unit roundoff. The key point is that the perturbation is small relative to the *data*, not relative to the answer.

We ask the same question of the DC algorithm. A DC function is $F = G - H$ with $G, H$ convex. One DCA step linearizes $H$ at the current iterate $x$ — form a subgradient $s \in \partial H(x)$ — and minimizes the *model* $M(z \mid x) = G(z) - \langle s, z\rangle$ to get $x_+$. In floating point both operations are rounded: the computer returns $\tilde{s} = \mathrm{fl}(s) = s + e$ (the *tilt* error $e$) and an approximate minimizer $\tilde{x}_+$ of $\tilde{M}(z \mid x) = G(z) - \langle \tilde{s}, z\rangle$.

The naive forward-error view gives poor bounds. The decomposition conditioning $\chi_{DC} = (G + H)/|F|$ can be doubly exponential in depth (thesis, Theorem 5.5.4), so evaluating $F = G - H$ suffers catastrophic cancellation. Yet DCA converges in practice, often in 1–2 steps, despite $\chi$ reaching $10^{16}$ [Observed, Unreproduced]. The backward view resolves the paradox: the *step* can be conditionally backward-stable even when the *evaluation* has large forward error.

The key is the gauge change. Adding the same convex function $C$ to both components, $(G, H) \mapsto (G + C, H + C)$, leaves $F = G - H$ unchanged but adds the Bregman divergence $D_C(y, x) = C(y) - C(x) - \langle \nabla C(x), y - x\rangle$ to the DCA majorant (thesis, Proposition 5.5.1) [Standard]. Geometrically this is a *gauge arrow* in the category $\mathcal{DC}^{\mathrm{gauge}}$ of DC decompositions; gauge arrows compose by addition.

**Remark 1.2** (The gauge category, informally)**.** The word "category" is used here in the working sense, not as a new formal definition: objects are DC pairs $(G, H)$; morphisms are gauge arrows $(G, H) \to (G + C, H + C)$ for convex $C$, composed by addition of gauges; a *component* is an equivalence class under gauge moves, i.e. the set of decompositions reachable from a given pair by adding common convex gauges.

**Remark 1.3** (Genealogy of the term "gauge")**.** The terminology of *gauge* for a common convex perturbation $(G, H) \mapsto (G + C, H + C)$ is due to Paper A; the thesis proves the same move as Proposition 5.5.1. We warn the reader that "gauge" here has nothing to do with gauge theory in physics: it means a change of decomposition that fixes the objective.

The target property needs a name before the gauge that witnesses it is constructed. To be precise, we need the following notion: a computed step $\tilde{x}_+$ is *backward-exact* if it equals the exact DCA step from $x$ of a gauge-perturbed pair $(G + C, H + C)$ for some convex $C$.

Since the tilt error $e$ must be absorbed by the gauge itself (the residual protocol of §3 will show it cannot simply be dropped), we include the linear tilt in the gauge. For $\sigma \geq 0$ define the *gauge*

$$C_{e,\sigma}(z) = \langle e, z\rangle + \tfrac{\sigma}{2}\|z - x\|^2.$$

**Remark 1.4** (Instantiation)**.** Explicitly, the gauged model is $M_{G+C,H+C}(z \mid x) = M(z \mid x) + \frac{\sigma}{2}\|z - x\|^2$: the linear tilt $\langle e, z\rangle$ cancels in the Bregman divergence, and only the quadratic part survives. The example to keep in mind is $G(z) = z^2/2$, $H = 0$ — the tilted subproblem of the erratum in §3. The condition $\sigma \geq 0$ is not needed for the identity of Theorem 1.5(a), but will be essential in §3, where it makes every fitted gauge convex and lets the PSD certificate apply.

**Theorem 1.5** (Backward exactness, *Standard* for (a) (thesis Proposition 5.5.1 [13]), *Conditional* for (b))**.** *Let $C = C_{e,\sigma}$ with $\sigma \geq 0$. Then:*

(a) *$C$ is convex, $\nabla(H + C)(x) = \tilde{s}$, and the gauged model satisfies $M_{G+C,H+C}(z \mid x) = M_{G,H}(z \mid x) + \frac{\sigma}{2}\|z - x\|^2$ (the linear part cancels in $D_C$).*

(b) If $x_+(\sigma^*) = \arg\min_z M(z \mid x) + \frac{\sigma^*}{2}\|z - x\|^2$ *equals the computed $\tilde{x}_+$, then $\tilde{x}_+$ is* backward-exact *from $x$ for the pair $(G + C_{e,\sigma^*},\ H + C_{e,\sigma^*})$.*

Part (a) is the gauge-change identity; it is unconditional. Part (b) is conditional: it reduces the slogan to a *feasibility* question — does a $\sigma^* \geq 0$ exist with $\|x_+(\sigma^*) - \tilde{x}_+\|$ small? — and feasibility can fail for algebraic reasons, not just numerical ones.

**Remark 1.6** (Forward reference)**.** Paper A §8 (the backward-error thread) uses this theorem: part (a) is the gauge identity used there, and part (b) is the conditional exactness whose feasibility §3 certifies. Through Paper A it feeds hub gap G5.

**Remark 1.7** (*Erratum* to v0.1: the obstruction)**.** The example to keep in mind is the following. Unconditional backward exactness is *false*. Take $G(z) = z^2/2$, $H = 0$, current iterate $x = 0$. Every convex differentiable common gauge $C$ gives the subproblem $\frac{1}{2}z^2 + D_C(z, 0)$, whose unique minimizer is $z = 0$. Hence *any* computed approximate minimizer $\delta \neq 0$, however small, is *not* backward-exact for *any* common convex gauge. This is an algebraic obstruction, not a grid-resolution artifact. (Counterexample supplied by external review.) Consequently every "every step is backward-exact" claim in v0.1 is weakened to conditional form in this version; the original wording is retained in the v0.1 file for the record.

**Remark 1.8** (Perturbed *problem* vs perturbed *algorithm*)**.** We distinguish our claim from the nearest "exact on perturbed $X$" result for any first-order method: Vernimmen and Glineur [9] show that worst-case inexact gradient descent equals *exact* gradient descent with a miscalibrated stepsize $h_{\text{eff}} = h(1\pm\delta)$ — exact on a perturbed *algorithm* (the stepsize), the function being minimized unchanged. Our claim is the reverse: the *computed* DCA step is *backward-exact* on a perturbed *decomposition* $(G + C, H + C)$, with the objective $F$ itself untouched. The two perturbation classes are disjoint; neither implies the other.

The non-obvious finding is the *validity range* of the slogan. Two separate roundings happen per step. The *tilt* $e = \tilde{s} - s$ is $O(u)$ [Observed, Unreproduced]. The *solve* is the fit above. The *rounding-noise scale* of a subproblem is $u\|P\|$: unit roundoff times the scale of the evaluated part.

The DCA step's backward-validity range is $u\|P\| \lesssim dM$, where $dM := M(x) - M(\tilde{x}_+)$ is the model's decrease at the computed step. Here $P$ is the function evaluated in floating point inside the subproblem — in the thesis splitting, $P = G$ itself — and $\|P\|$ is its magnitude scale. This is *not* $\chi u$. The conditioning $\chi$ governs *evaluating* $F = G - H$ (forward, thesis Lemma 5.5.3); the quantity $u\|P\|/dM$ governs the *step* (backward). They differ by $|F|/dM$.

We state the criterion $u\|P\| \lesssim dM$ as *observed* — a signal-to-noise criterion, not a universal theorem: it depends on the representation and the solver, and adding a common constant changes $\|P\|$ while leaving the exact step unchanged.

Unlike linear-system solves, DCA's backward stability is *conditional*: its validity range is set by the subproblem's signal-to-noise ratio, and $\chi$ enters the data ($P$'s absolute rounding-noise scale is $u\|P\| \sim u\chi|F|$). Within the validity range, the fitted gauges change $\chi$ only slightly: the fitted $C_e$ (the per-step fitted gauge $C_{e,\sigma}$ of §1) changes $\chi$ by $2C_e/|F| \ll \chi_0$ (with $\chi_0$ the initial conditioning, away from $F = 0$), so backward stability does not improve conditioning — which is Wilkinson's point, now stated precisely for DCA.

**Remark 1.9** (Why naive DCA converges without reliable $F$-values)**.** This explains an old puzzle [Observed]: naive DCA converges in 1–2 steps despite $\chi \sim 10^{16}$ (steps are backward-exact *when feasible*), but it cannot evaluate $F$ reliably (the $F$-evaluation has large forward error). The two phenomena belong to different error analyses.

The Lean formalization of descent with gauge machine-checks the companion identity $(G - H)(x_+) - (G - H)(x) \le D_h(x_+, x) - D_C(x_+, x)$ with zero `sorry`.

**Remark 1.10** (Service statement)**.** This paper serves Paper A (the backward-error thread, §8 there) and, through it, the hub paper (the DCA–sheaf

thread, gap G5 on sheaf–DCA functoriality). Every main result below carries a forward reference to the thread it feeds.

## 2. Three regimes

The purpose of this section is to exhibit the three observed regimes, to locate the validity boundary of the conditional criterion, and to show where the quantitative slogan weakens. We note at this stage that the boundary is an empirical matter; the protocol is in the appendix.

The three regimes give three different answers to the feasibility question of §1. In Regime A the slogan holds at solver tolerance; in Regime B it weakens — a backward gauge exists, but it is large; in Regime C no gauge explains the steps at all. The reader should compare each regime with the validity range $u\|P\| \lesssim dM$ of §1: above it, the slogan holds; at its edge, it degrades; beyond it, fit is impossible.

On the ReLU chain (thesis §5.18) with Powell subproblems in float32, depth $L$ drives the experiment through three regimes [Computed, Reproduced] (packet `reproduction/`, `exp1_relu_chain.py`, seed 0):
Regime A (solver solves). $\sigma$-match error $10^{-4}$–$10^{-3}$ (the distance $\|x_+(\sigma^*) - \tilde{x}_+\|$ between the fitted gauge's exact step and the computed step) with small or zero $\sigma^*$: the slogan holds at *solver tolerance*, not at rounding-noise scale. Powell's tolerance $10^{-6}$ is not backward-stable at the $O(u)$ scale; derivative-free methods leave $O(\text{tol})$ gradient residual.
Regime B (kink stall). Powell stops at kinks after one iteration; explaining the micro-step (a kink-stall step with tiny $dM_k$, the per-step model decrease) needs $\sigma^* \sim 10^5$–$10^7$ with $D_C/dM \gg 1$: the backward gauge is *large* and the quantitative slogan weakens.

**Remark 2.1** (The kink-stall scaling law and bound; *[Computed]*)**.** Write $d := \tilde{x}_+ - x$ for the computed step and $w := D_C/dM$ for the gauge-work share. The kink-stall steps were re-run with the step length $\|d\|$, the model decrease $dM$, and Powell's iteration count measured independently (packet `reproduction/`, `exp1_relu_chain.py`, `analyze1.py`, `verify_bound.py`; LB-764–LB-769). On every kink-stall step — Powell stops after exactly one iteration, $dM > 0$, and the $\sigma$-match is good — the product $\sigma^* \cdot \|d\|$ is constant, independent of $\|d\|$: at depth $L = 5$, $\sigma^* \cdot \|d\| = 0.86 \pm 0.06$ on 7 steps, and across 33 stall steps the median is 0.78, with two-thirds of the steps in $[0.3, 3]$. Call this calibrated constant $\rho_{\text{fit}}$: the residual scale of the fit, at the signal scale $O(1)$. This gives a closed-form bound, calibrated once per (problem, depth):

$$\sigma^*_{\text{pred}} = \rho_{\text{fit}}/\|d\|, \qquad D_{C,\text{pred}} = \rho_{\text{fit}}\|d\|/2, \qquad w_{\text{pred}} = D_{C,\text{pred}}/dM = \rho_{\text{fit}}\|d\|/(2\,dM).$$

The bound was tested on 15 *new* kink-stall steps ($L = 5$, six trajectories, fresh seeds; $\rho_{\text{fit}} = 0.862$ calibrated on the first 7 steps, not on the new ones): on true micro-steps ($\|d\| < 0.25$) the median relative error is 0.06, max 0.15. It does not apply when $\|d\| > 0.3$: such steps are no longer stalls (Regime A

scale) and the error is large, as expected. The mechanism (LB-769) is that a large $\sigma^*$ is *not* caused by a large residual — the residual of the fit stays $O(1)$ — but by a micro step length: $\sigma^* = \rho_{\text{fit}}/\|d\|$. The $\sigma$-match implicitly selects the correct subgradient in $\partial G(\tilde{x}_+)$. The one-dimensional toy $M(z) = |z|$ is solvable in closed form: $\sigma^* = \rho/\|d\|$ exactly, with $w = 1/2$; $w \gg 1$ needs $dM \ll \rho_{\text{fit}}\|d\|/2$ (severe, sub-first-order stall), which does not come for free. Limits: $\rho_{\text{fit}}$ is calibrated at $L = 5$ only; other depths are open. The heaviest stalls (large $\sigma^*$ with poor $\sigma$-match) need finer experiments.

**Theorem 2.2** (No exact isotropic gauge for an uphill step; *Proved — agent level; pending author review; Lean verification pending.*)**.** *Let $\tilde{x}_+$ be a computed step with $dM := M(x) - M(\tilde{x}_+) < 0$ (the model* increases *at the computed step). Then for every $\sigma \geq 0$ the regularized minimizer $x_+(\sigma) = \arg\min_z M(z \mid x) + \frac{\sigma}{2}\|z - x\|^2$ satisfies $x_+(\sigma) \neq \tilde{x}_+$.* Proof. *$M_\sigma(\tilde{x}_+) = M(\tilde{x}_+) + \sigma\|d\|^2/2 > M(x) = M_\sigma(x)$, since $M(\tilde{x}_+) > M(x)$ and the $\sigma$-term is nonnegative, so $\tilde{x}_+$ is never the minimizer.* □

**Remark 2.3** (The uphill theorem explains the "no gauge" structure)**.** Theorem 2.2 (LB-767) is the structural reason behind the "no gauge" of Regime C and severe stall: an uphill step is never backward-exact for any isotropic gauge — this is not a limitation of the $\sigma$-grid.

**Remark 2.4** (Warning)**.** We warn the reader that "small gauge" is not part of the definition of backward exactness: in Regime B the explaining gauge has $\sigma^* \sim 10^5$–$10^7$. Backward exactness is a structural property (existence of some feasible $C$); smallness is the separate quantitative question answered by the size bound of §3.

Regime C ($\chi u_{32} \gg 1$, $L = 6$). With $u_{32} = 2^{-24}$ the float32 unit roundoff, float32 DCA diverges; $\sigma$-match error $\sim 1$: *no* gauge explains the steps. The float32 subproblem ($P \sim 2 \times 10^{11}$, rounding-noise scale $\sim 2 \times 10^4 \gg$ progress) is dominated by rounding noise. This is the sharp boundary of the conditional criterion: fit becomes impossible exactly when $u\|P\| \gg dM$ [Observed, Reproduced].

Regime C locates the boundary empirically: fit fails exactly when the rounding noise exceeds the step's progress. The next section proves the quantitative content above the boundary, in one dimension.

## 3. The composition bound in one dimension

The purpose of this section is to fix the residual protocol, to prove the per-step size bound, and to give the PSD feasibility certificate. §§3–4 assume $H$ differentiable at $x$ ($s = \nabla H(x)$); the nonsmooth case (§2, Regime B at kinks) is outside this analysis. Since a simple exact-fit algebra is available in one dimension, we work there; the general-$n$ alignment estimate is in §4.

**Convention status (§3 vs §4).** In v0.5–v0.7, §3 was internally inconsistent on the fit equation: two prose sentences stated it as $Qd = r - e$ ("not $Qd = r$"), while all the mathematics of §3 — Theorem 3.3 with $\sigma_k^* = r_k/d_k$, the PSD certificate with $Qd = r$, and the certificate definition with $d_k^T r_k > 0$ — used $Qd = r$. This is resolved in v0.8: the prose is unified on $Qd = r$ (see Remark 3.2). §4 (Theorem 4.1) uses $Qd = r$ throughout.

We fix notation for the whole section. In §1's notation, $s^e = \tilde{s}$ and $y = \tilde{x}_+$ (both computed); $x_+(\sigma)$ below denotes the exact minimizer of the $\sigma$-regularized model, while $x_{k+1}$ denotes the computed iterate. Since the tilt error $e$ will turn out to be non-droppable, every symbol records whether it is rounded or not. Write $s = \nabla H(x)$ for the *original* (unrounded) slope, $s^e = s + e$ for the computed slope, $y$ for the computed minimizer, and $d = y - x$ for the step. The gauge to be fitted is

$$C(z) = \langle e, z\rangle + \tfrac{1}{2}(z - x)^T Q(z - x), \qquad Q \succeq 0.$$

Exact gauged stationarity is $\nabla G(y) + Qd = s$. Minimizing the gauged model gives $\nabla G(y) + \nabla C(y) = \tilde{s}$; with $\nabla C(y) = e + Qd$ and $\tilde{s} = s + e$ the tilt cancels. Define the residual against the *original* slope,

$$r = s - \nabla G(y),$$

so that the fit equation is $Qd = r$: the tilt cancels between $\nabla C(y) = e + Qd$ and $\tilde{s} = s + e$. The tilt error $e$ cannot be dropped from the protocol, however — the residual must be formed against the original slope $s$, not the computed $s^e$ (the v0.1 error corrected in the erratum below).

**Remark 3.1** (*Erratum* to v0.1: the tilt)**.** v0.1 set $r = s^e - \nabla G(y)$ and fitted $Qd = r$. The tilt example shows this is wrong. Take $G = z^2/2$, $H = 0$, $x = 1$, $s^e = e = 1/10$, $y = 1/10$ (the exact minimizer of the tilted subproblem). The naive formula gives $r = s^e - G'(y) = 0$, i.e. $\sigma = 0$ — but gauged stationarity $G'(y) + Qd = s$ still has residual $1/10$. The correct common quadratic gauge has $\sigma = 1/9$: indeed $1/10 + \sigma(1/10 - 1) = 0$ gives $\sigma = 1/9$. (Example supplied by external review.) All quantitative fits in this version use $r = s - \nabla G(y)$ and $Qd = r$; the v0.1 numbers that used the naive $r$ are marked *[Unreproduced]* pending recomputation.

**Remark 3.2** (*Erratum* (v0.5, packaging): the fit equation)**.** In v0.5–v0.7 the prose of §3 stated the fit equation as $Qd = r - e$, "not $Qd = r$", while all the mathematics used $Qd = r$. Under the definitions in force this was inconsistent: exact gauged stationarity is $\nabla G(y) + Qd = s$, and with the corrected residual $r := s - \nabla G(y)$ this gives $Qd = r$; the tilt example above confirms it ($1/10 + \sigma(1/10 - 1) = 0$ gives $\sigma = 1/9$, i.e. $Qd = r$). Corrected in v0.8: the prose is unified on $Qd = r$; §4 uses this convention throughout.

In one dimension put $\sigma_k^* = \max(r_k/d_k, 0)$ and $C_k(z) = \langle e_k, z\rangle + \frac{\sigma_k^*}{2}(z - x_k)^2$; when $r_k d_k < 0$ the max gives $\sigma_k^* = 0$ — the fallback, an approximate rather than exact fit.

The bound below uses two auxiliary symbols whose roles we record here: $s_r$ is the residual slope scale (the $|s_r|$ term in $S_k$ measures the tilt-error contribution to the roundoff scale), and $f_k$ is the Newton residual at the computed iterate, controlled by the stagnation estimate $|f_k| \le C_N u |x_{k+1}| G''(x_{k+1})$ stated after the theorem. The 1D scale $s_r$ is the one-dimensional counterpart of $\|s\|$ in the $n$-dimensional roundoff analogy of Theorem 4.1 ("On $\rho$ and $c$").

**Theorem 3.3** (Per-step bound, *Conditional*, 1D double-well, modulo Newton-stagnation estimate)**.** *If $r_k d_k > 0$ then $x_+(\sigma_k^*) = x_{k+1}$ exactly (stationarity plus strict convexity), so $\sigma_k^* = r_k/d_k$ and $D_{C_k}(x_{k+1}, x_k) = r_k d_k/2$. In general $D_{C_k}(x_{k+1}, x_k) = \max(r_k d_k, 0)/2 \le (uS_k|d_k|)/2$ with $S_k = 1.77|s_r| + 1.32|x_{k+1}|G''(x_{k+1})$, measured $\max_k |r_k|/(uS_k) = 0.70 \le 1$ [Reproduced] (reproduction: $0.49 \le 1$; the inequality is the claim). Summing,*

$$R_K := \sum_{k<K} D_{C_k} \le \tfrac{u}{2}\Big(\max_{k<K} S_k\Big)\,\mathrm{TV}_K, \qquad \mathrm{TV}_K = \sum_{k<K} |d_k|,$$

*measured $R_K = 2.79 \times 10^{-8}$ against the bound $8.83 \times 10^{-7}$ (per-step slack 2.6×) [Reproduced] (reproduction: $R_K = 6.10 \times 10^{-8}$, same order; the bound matches exactly at $8.8319 \times 10^{-7}$ and stays slack; per-step slack 8.0× by Σ-ratio, max 36.8, median 5.9). See Remark 3.4.*

The bound is modulo the Newton-stagnation estimate $|f_k| \le C_N u |x_{k+1}| G''(x_{k+1})$ (proved in "On $\rho$ and $c$" below; Higham's $\gamma_m$ analysis proves the *form* of the estimate and gives valid constants, but they come out about 6× looser where it matters ($c_1^{\mathrm{anal}} \approx 11$ against the measured 1.77) and they depend on implementation details (operation counts, the solver's backward-error constants), so there is no universal analytic replacement. The measured constants stand, certified by the ratio test described there) and is stated in one dimension, where the exact-fit algebra is vacuous — the content is the *size*, now proved. The composable quantity is $D_{C_k}$, not $\sigma_k^*$: on $d_k \to 0$ steps $\sigma^* = r_k/d_k$ is ill-behaved while $D_k = r_k d_k/2 \to 0$ continuously, including the fixed-point case $d_k = 0$.

**Remark 3.4** (Reproduction of the Exp-2 numbers (W5))**.** The Exp-2 numbers above were re-implemented from the protocol text and re-run (packet `reproduction/`: `exp2_doublewell.py`, `exp2_data.json`, `exp2_summary.json`; numpy 1.26.4, scipy 1.11.4, seed 0, no RNG used; LB-752–LB-755). The composition bound matches exactly ($8.8319 \times 10^{-7}$); the residual scale satisfies $\max_k |r_k|/(uS_k) = 0.49 \le 1$; the $\sigma$-match arbiter confirms the algebraic fit equation $Qd = r$ to max relative disagreement $1.5 \times 10^{-8}$; $\bar{\chi} = 9.12$ against the paper's 9.16. The reproduced $R_K = 6.10 \times 10^{-8}$ is the same order as the paper's $2.79 \times 10^{-8}$, and the bound stays slack. The constants 1.77/1.32 in $S_k$ are measured constants from the original run, reused as printed (the paper is explicit about this). The solver-sensitive tail statistics (feasibility ratio, gauge-work margin, curvature share) did *not* reproduce under a

protocol-faithful Newton and are reported with both values in §5; the pinned stopping rule for future runs is in the appendix protocol.

**Remark 3.5** (Forward reference)**.** Paper A §8 uses this bound: the composable quantity $D_{C_k}$ is the trajectory-level size control for the backward-error thread there.

**Theorem 3.6** (PSD feasibility certificate; proof supplied by external review)**.** *Let $d \neq 0$ and $r$ be as above. A positive semidefinite quadratic gauge with $Qd = r$ exists iff $d^T r > 0$ (or $r = 0$, with $Q = 0$).*

*Proof supplied by external review.* If $d^T r > 0$, put $Q = rr^T/(d^T r)$; then $Q \succeq 0$ and $Qd = r(r^T d)/(d^T r) = r$. If $r = 0$, $Q = 0$ works. Conversely, if $Q \succeq 0$ and $d^T Q d = 0$ then $Qd = 0$; hence $d^T r = 0$, $r \neq 0$ admits no PSD quadratic fit. The isotropic ansatz $Q = \sigma I$ needs the stronger condition $r \parallel d$. Moreover $D_C(y, x) = \frac{1}{2} d^T r \leq \frac{1}{2}\|r\|\|d\|$, $\|Q\| = \|r\|^2/(d^T r)$; keeping the gauge curvature small requires controlling the angle $d^T r/(\|d\|\|r\|)$, since small work alone does not suffice. □

**Remark 3.7** (Scope of the Lean check)**.** The constructive half above (existence of the PSD fit via $Q = rr^T/(d^T r)$) is machine-checked in Lean (`psd_fit_norm`, T02); the necessity half is the hand proof above, supplied by external review.

**Definition 3.8** (Conditional backward-error certificate)**.** A fitted gauge $C_k$ satisfying the PSD feasibility condition $d_k^T r_k > 0$ above, together with the size bound of Theorem 3.3, is a *conditional backward-error certificate* for step $k$: it certifies the step as backward-exact on the gauge-perturbed pair, conditionally on feasibility.

**Remark 3.9** (Instantiation)**.** In the double-well run below, the certificate for a progressing step with $r_k d_k > 0$ is $C_k(z) = \langle e_k, z\rangle + \frac{\sigma_k^*}{2}(z - x_k)^2$ with $\sigma_k^* = r_k/d_k$; the 84% of progressing steps with $r_k d_k < 0$ admit no PSD quadratic certificate at all.

**Remark 3.10** (Forward reference)**.** Paper A §8 uses the feasibility certificate as the per-step existence check for a conditional backward-error certificate.

On the double-well run, exact-fit feasibility ($r_k d_k \geq 0$) holds on only 16% of progressing steps ($d_k \neq 0$) [Solver-sensitive]; the other 84% admit no PSD quadratic certificate (Remark 3.11). On stagnation steps ($d_k = 0$) it never holds with strict inequality [Unreproduced].

**Remark 3.11** (Trace of the 16%/84% definition)**.** The 16% figure counts progressing steps ($d_k \neq 0$) with $r_k d_k \geq 0$ under the *corrected* residual $r = s - \nabla G(y)$; the denominator is all progressing steps. This definition must be synced with Paper A §8 (p. 13), which reports a feasibility ratio under the same name; until the two protocols are unified the numbers are not comparable.

**Remark 3.12** (Solver sensitivity of the 16%/84% split)**.** The 16%/84% split depends on the Newton stopping rule of the lost original script: the float32 DCA map near the fixed point $x = 1$ has an attracting 2-cycle ($1.0 \leftrightarrow 0.9999999404$, one ulp), so whether the run stagnates ($d = 0$, as the paper's text implies) or cycles depends on a single ulp in the Newton stop (LB-753). The protocol-faithful reproduction (packet `reproduction/`) cycles and gives 10% (15/150) feasibility; cutting the reproduced run at $k < 20$ (as if the original stagnated there) recovers a gauge-work margin and a curvature share within 1.7× of the paper's. The hypothesis that "16% is legacy of the naive residual" is ruled out. The paper keeps the original run's value; the reproduction value is reported alongside, not substituted.

The size bound of Theorem 3.3 deliberately does not need feasibility. Part (b) of Theorem 1.5 is backward-exact only rarely; the approximate form ($\sigma^* = 0$ fallback) is the norm.

## 4. The alignment estimate in $n$ dimensions

The purpose of this section is to state and prove the alignment estimate that lifts the quantitative gauge-fit bounds of §3 from one dimension to general $n$. In one dimension the fit equation is scalar and the exact-fit algebra is vacuous; in $n > 1$ the fitted gauge $Q \succeq 0$ must align the residual $r$ with the step $d$, and the size of the explaining gauge is controlled by the angle between them. The estimate is conditional on the PSD feasibility certificate of §3, exactly as the 1D bound is conditional on $r_k d_k > 0$. We note at this stage that the size of the gauge and its curvature are distinct quantities, and that the alignment $\cos\theta$ controls the curvature but not the fit coefficient.

**Theorem 4.1** (The alignment estimate in $n$ dimensions; *Proved — agent level; pending author review; Lean verification pending.*)**.** *Convention for this section. $r := s - \nabla G(y)$ is the corrected residual of §3 and the fit equation is $Qd = r$ throughout; see Remark 3.2 for the v0.5–v0.7 erratum history. Fix a step with $d := y - x \neq 0$, $\theta$ the angle between $d$ and $r$, $c := \cos\theta = d^T r/(\|d\|\|r\|)$, and $\rho \geq \|r\|$ the roundoff noise scale.*

*(a)* Fit coefficient *[Proved — agent level]. The constrained least-squares fit $\sigma_{\mathrm{LS}} := \arg\min_{\sigma \geq 0} \|\sigma d - r\|$ is $\sigma_{\mathrm{LS}} = \max(d^T r/\|d\|^2, 0)$, with misfit $\|r - \sigma_{\mathrm{LS}} d\|^2 = \|r\|^2 - \max(d^T r, 0)^2/\|d\|^2$ — in the acute case $d^T r \geq 0$ this is $\|r - \sigma_{\mathrm{LS}} d\| = \|r\| \sin\theta$; exact fit $\sigma d = r$ with $\sigma \geq 0$ is solvable iff $r \parallel d$ and $d^T r \geq 0$, and then $\sigma^* = d^T r/\|d\|^2$ (the direct $n$-dimensional analogue of $r/d$). In all cases $|\sigma_{\mathrm{fit}}| \leq \rho/\|d\|$, with no alignment condition.*

*(b)* Curvature of the PSD fit *[Proved — agent level; conditional on $d^T r > 0$]. For the rank-one certificate $Q = rr^T/(d^T r) \succeq 0$ of the PSD feasibility certificate,*

$$\|Q\| = \frac{\|r\|^2}{d^T r} = \frac{\|r\|/\|d\|}{\cos\theta} \lesssim \frac{\rho}{c\,\|d\|}, \qquad c \in (0, 1],$$

*and the smallest isotropic gauge dominating $Q$ ($\sigma I \succeq Q$) is $\sigma_{\mathrm{dom}} = \|Q\|$, satisfying the same bound. This is the $n$-dimensional content of the conjectured form $\sigma^* \lesssim \rho/(c\|d\|)$: the denominator $c$ belongs to the* curvature*, not to the fit coefficient of (a).*

*(c)* Work–curvature trade-off *[Proved — agent level]. With $D_C := \frac{1}{2} d^T r$,*

$$D_C \cdot \|Q\| = \frac{\|r\|^2}{2}.$$

*At fixed residual scale $\|r\|$, small gauge work is equivalent to large curvature; the two trade through $\cos\theta$.*

*(d)* Composability *[Proved — agent level; pending author review; Lean verification pending.]. Put $D_{C_k} := \frac{1}{2}\max(d_k^T r_k, 0)$. Then*

$$R_K := \sum_{k<K} D_{C_k} \le \frac{u}{2}\Big(\max_{k<K} S_k\Big)\,\mathrm{TV}_K, \qquad \mathrm{TV}_K := \sum_{k<K} \|d_k\|,$$

*with no alignment hypothesis and no feasibility hypothesis (the roundoff estimate brings in $\|\nabla^2 G\|$ through $S_k$): the size part of Theorem 3.3 lifts to $n$ dimensions intact, resting on the roundoff estimate of "On $\rho$ and $c$" below (same agent-level label).*

*On $\rho$ and $c$. The $n$-dimensional roundoff estimate $\|r_k\| \lesssim u S_k$ with $S_k = c_1\|s\| + c_2\|y_k\|\,\|\nabla^2 G(y_k)\|_{\mathrm{op}}$ is [Proved — agent level; pending author review; Lean verification pending.]. Here $u$ is the unit roundoff ($u = 2^{-24}$ in float32), $s = \nabla H(x_k)$ is the exact slope, $y_k$ is the computed subproblem minimizer, and $\|\cdot\|_{\mathrm{op}}$ is the spectral (operator) norm. The proof (full details in* `agent-notes/ROUNDING-ND-PROOF.md`*) splits the corrected residual $r_k = s - \nabla G(y_k)$ into two parts: $r_k = -e - g$, where $e = \tilde{s} - s$ is the tilt error of the computed slope $\tilde{s} = fl(\nabla H(x_k))$ (pure evaluation rounding) and $g = \nabla G(y_k) - \tilde{s}$ is the subproblem gradient at the computed minimizer. The tilt part satisfies $\|e\| \le u\, c_1\|s\|$; the solver part satisfies $\|g\| \le u\, c_2\|y_k\|\,\|\nabla^2 G(y_k)\|_{\mathrm{op}}$ from the solver's stopping rule, with no condition number of the Hessian entering — only backward stability of the linear solve is used. The estimate holds under an explicit assumption set: $G$ is $C^2$ with $\nabla^2 G \succ 0$ on the trajectory (smooth steps only); the Newton solver stops by one of four named rules — exact float32 fixed point, relative stagnation ($\|\hat{p}\| \le 2u\|y\|$ with $\hat{p}$ the computed correction), exactly-zero computed gradient, or gradient-stall. For gradient-stall we assume stall-faithfulness: the computed gradient at a stall is dominated by evaluation noise, so $\|g\| \lesssim u\|s\|$. This is a standing backup hypothesis: the gradient-stall path never fired in the 1500-step 2D probe, so it has no numerical verification yet. The constants $c_1, c_2$ are measured per (problem, code) and certified by the ratio protocol ($\max_k \|r_k\|/(uS_k) \le 1$; the final Newton damping factor is at most 2, absorbed into $c_2$). At kinks (Regime B) $\nabla^2 G$ does not exist and the estimate does not apply; nonsmooth steps need a separate analysis. The alignment $c$ is a per-step measured quantity; no lower bound for $c$ follows from the Hessian spectrum alone.*

The $S_k$ here is the $n$-dimensional analogue of the 1D roundoff scale $S_k$ of Theorem 3.3; $c_1, c_2$ are the corresponding measured constants, certified by the ratio protocol (tight analytic values [Open]; the Higham $\gamma_m$ analysis justifies the form). The norm on $\nabla^2 G(y_k)$ in $S_k$ is the spectral (operator) norm.

In outline: (a) the least-squares problem is a scalar parabola in $\sigma$, so the fitted coefficient $\sigma_{\text{fit}}$ (the constrained minimizer) is $\max(\cdot, 0)$; (b) $Q$ is rank one, hence has a single nonzero eigenvalue, which sets the curvature; (c) gauge work $D_C$ and curvature $\|Q\|$ are linked through $\cos\theta$; (d) the size bound composes because no step of it uses $\cos\theta$, the Hessian, or feasibility.

The proof has four independent parts, one for each item of the theorem.

*Proof.* (a) $\|\sigma d - r\|^2 = \sigma^2\|d\|^2 - 2\sigma d^T r + \|r\|^2$ is a parabola in $\sigma$ with unconstrained minimizer $\sigma = d^T r/\|d\|^2$; the convexity constraint $\sigma \geq 0$ gives $\max(\cdot, 0)$. If $d^T r \geq 0$ the minimizer is interior, $d^T(r - \sigma d) = 0$, and Pythagoras gives $\|r - \sigma d\|^2 = \|r\|^2 - (d^T r)^2/\|d\|^2 = \|r\|^2 \sin^2\theta$. If $d^T r < 0$ the constrained minimizer is at the boundary $\sigma = 0$, so $\|r - \sigma d\|^2 = \|r\|^2$. Together, $\|r - \sigma_{\text{LS}} d\|^2 = \|r\|^2 - \max(d^T r, 0)^2/\|d\|^2$, which is $\|r\|^2 \sin^2\theta$ in the acute case. Exact fit $\sigma d = r$ with $\sigma \geq 0$ forces $r \parallel d$ and $d^T r = \sigma\|d\|^2 \geq 0$, and then $\sigma = d^T r/\|d\|^2$; this reduces to $r/d$ in one dimension. Cauchy–Schwarz gives $|d^T r| \leq \|d\|\|r\| \leq \rho\|d\|$, whence the bound.

(b) $Q$ is rank one with eigenvector $r$: $Qr = r\,\|r\|^2/(d^T r)$, all other eigenvalues 0; with $d^T r > 0$ the operator norm is $\|Q\| = \|r\|^2/(d^T r) \geq 0$. Rewriting, $\|r\|^2/(d^T r) = (\|r\|/\|d\|)\cdot(\|r\|\|d\|/(d^T r)) = (\|r\|/\|d\|)/\cos\theta$, and $\|r\| \leq \rho$ gives the estimate. $\sigma I \succeq Q$ iff $\sigma \geq \lambda_{\max}(Q) = \|Q\|$.

(c) $D_C = \frac{1}{2} d^T Q d = \frac{1}{2}(d^T r)^2/(d^T r) = \frac{1}{2} d^T r$; multiplying by $\|Q\| = \|r\|^2/(d^T r)$ gives $\|r\|^2/2$.

(d) $D_{C_k} \leq \frac{1}{2}\|d_k\|\|r_k\|$ by Cauchy–Schwarz and $\|r_k\| \leq u S_k$ by the per-step roundoff bound, so $D_{C_k} \leq (uS_k/2)\|d_k\|$; summing over $k < K$ and bounding $S_k$ by its maximum gives the claim. No step uses $\cos\theta$, the Hessian, or the feasibility $d_k^T r_k > 0$ (the max covers the infeasible steps). □

**Remark 4.2** (*Erratum* to v0.5: the misfit identity)**.** In v0.5, part (a) of Theorem 4.1 stated the misfit identity without a case restriction: $\sigma_{\text{LS}} = \max(d^T r/\|d\|^2, 0)$ with misfit $\|r - \sigma_{\text{LS}} d\| = \|r\|\sin\theta$. This is false when $d^T r < 0$. Counterexample: $d = (1, 0)$, $r = (-1, 1)$. Then $\sigma_{\text{LS}} = 0$ and the true misfit is $\|r\| = \sqrt{2} \approx 1.414$, while the v0.5 formula gives $\|r\|\sin\theta = \sqrt{2}\sin 135^\circ = 1$. The v0.5 proof computed orthogonality at the unconstrained minimizer, which does not apply at the constrained minimizer $\sigma = 0$. The correct identity is $\|r - \sigma_{\text{LS}} d\|^2 = \|r\|^2 - \max(d^T r, 0)^2/\|d\|^2$; the sine form $\|r - \sigma_{\text{LS}} d\| = \|r\|\sin\theta$ holds only in the acute case $d^T r \geq 0$, as part (a) and its proof now state. The error was found during Lean-ready ticket preparation for v0.5 (2026-10-16) and corrected in v0.8 following the author's rule to enter the correct form directly. (b) and (c) assume $d^T r > 0$ and are unaffected; (d) uses the max-form $D_{C_k} := \frac{1}{2}\max(d_k^T r_k, 0)$ and needs

no such assumption. Lean verified the corrected statement of Theorem 4.1 (all four parts, 0 sorry; merged).

The numerical check below measures every quantity the theorem controls, in the simplest $n > 1$ setting where misalignment can occur: the 2D double-well.

**Remark 4.3** (Numerical check)**.** Protocol *[Computed]*: 2D double-well $F(x, y) = x^4 - 2x^2 + y^4 - 2y^2 + 0.3xy$ with an explicit convex DC splitting, DCA in float32, 10 starts × 150 steps (1500 steps, 323 progressing); Newton with backtracking on the gradient norm and true float32 stagnation (the solver halting when successive float32 iterates are identical). Corrected residual $r = s - \nabla G(y)$; rank-one certificate $Q = rr^T/(d^T r)$ on the feasible steps. Script: `e3r1_align2d.py` (agent-notes/code).

We set $w_k := D_{C_k}/dM_k$, the per-step gauge-work share (the fraction of the step's model decrease accounted for by the fitted gauge); the *gauge-dominance boundary* $w = 1$ is the threshold at which the fitted gauge accounts for all of the step's model progress.

| Quantity | Measured value |
|---|---|
| $\|Q\| = \|r\|/(\|d\|\cos\theta)$ identity (code sanity check) | holds to $10^{-9}$ on every feasible step |
| $\max_k \|r_k\|/(uS_k)$ with $u = 2^{-24}$ | 0.91 (median 0.31) |
| $\cos\theta$ on feasible steps | min 0.0079, Q1 0.35, median 0.67, Q3 0.93, max 1.0 |
| worst alignment penalty $1/\cos\theta$ | 126× |
| feasible steps $d^T r > 0$ | 203/323 = 62.8% of progressing steps |
| isotropic $\sigma^*_{\text{match}}$ (best-match $\sigma$) vs $d^T r/\|d\|^2$ | median relative deviation 3.7% |
| log-log slope of $\sigma^*$ vs $1/\|d\|$ | $1.035 \approx 1$ over 7 decades |
| micro-step outlier (one step) | $\|d\| = 1.3 \times 10^{-7}$: $\sigma^*_{\text{match}} = 1.11$, $\|Q\| = 587$, $D_C = 3.3 \times 10^{-16}$ (see Remark 4.4) |
| $R_K = \sum D_C$ against the composition bound | $2.19 \times 10^{-6}$, per-run slack 3–913× |
| $\max_k D_C/dM_k$ (gauge-work share) | 0.255 (median $5.7 \times 10^{-7}$), below the $w = 1$ boundary |
| stopped-short feasible steps | 173/203 *[Observed]* |

**Remark 4.4** (On the micro-step outlier row)**.** The $\sigma^*_{\text{match}} = 1.11$ is a line-search artifact on a flat valley (best misfit still 25% of $\|d\|$); the validated least-squares fit gives $\sigma_{\text{ls}} = 0.037$ (the least-squares coefficient), and $\|Q\| = \sigma_{\text{ls}}/\cos^2\theta = 587.3$ matches exactly. The conclusion — the isotropic gauge cannot fit a strongly misaligned step — follows from the 25% misfit.

The stopped-short mechanism (float32 Newton halting before the tilted minimizer, tilting $r$ toward $d$) is a measured tendency with a proposed mechanism, *[Observed]*, not a theorem; the uniform worst case $c_{\min} = 0.0079$ is measured, not a proved constant. All numbers in the table are *[Computed, Unreproduced]* until the scripts enter the accompanying materials (see the appendix protocol).

**Remark 4.5** (Forward reference)**.** Paper A §8 uses this estimate: it is the $n$-dimensional content of item (d) of the backward-error theorem there, lifting the composition bound from 1D.

## 5. The formal gauge sequence

The purpose of this section is to define the formal cumulative gauge, to record the measurements on the gauge-fit sequence, and to separate both from a realized walk.

Terminology for this section: the *gauge-fit sequence* is the per-step fitted gauges $\{C_k\}$; the *formal cumulative gauge* is $S_K = \sum_{k<K} C_k$ (uppercase $K$; not to be confused with the per-step roundoff scale $S_k$ of §§3–4). A *realized walk* would be a whole run whose steps are DCA steps of the cumulative pairs — this is open (see the erratum below), so "walk" below always means the formal object unless marked realized.

The following structural claim is *[Theorem]*: every $C_k$ is convex by construction ($\sigma_k^* \geq 0$), so $S_K = \sum_{k<K} C_k$ is convex and $(G + S_K, H + S_K)$ lies in the gauge component of $(G, H)$. The formal cumulative gauge cannot jump components — this follows from the construction.

The key point is that, quantitatively [Computed, Solver-sensitive], the gauge-fit sequence stays far from the boundary. The per-step gauge-work share $w_k = D_{C_k}/dM_k$ measures the fraction of the step's model decrease accounted for by the fitted gauge; the *gauge-dominance boundary* is $w = 1$ (the kink-stall / deliberate-jump threshold): at $w = 1$ the fitted gauge accounts for all of the step's model progress.

On all progressing steps $\max_k w_k = 8.87 \times 10^{-4}$: the sequence stays $\geq 1127\times$ *below the boundary, uniformly* [Solver-sensitive] (reproduction: $\max_k w_k = 0.144$, about $7\times$ below the boundary; see Remark 5.1).

A second metric: gauge curvature $< 0.2\%$ of problem curvature ($\max \sigma_k^* / \min G'' \approx 1.9 \times 10^{-3}$, with $\min G''$ over the trajectory) [Solver-sensitive] (reproduction: 0.43; see Remark 5.1). The fitted gauges change $\chi$ only slightly: they change it by $< 2 \times 10^{-8}$ relatively ($\sum 2D_{C_k}/|F_k| = 1.59 \times 10^{-7}$ against $\bar{\chi} \approx 9.16$, the run-average $\chi$) [Reproduced] (reproduction: $4.42 \times 10^{-7}$ against $\bar{\chi} = 9.12$).

**Remark 5.1** (Solver-sensitive tail statistics)**.** The margin numbers above ($\max_k w_k$, the curvature share) and the 16%/84% feasibility split of §3 depend on the Newton stopping rule of the lost original script and did not reproduce under a protocol-faithful Newton (packet `reproduction/`: $\max_k w_k = 0.144$, curvature share 0.43, feasibility 10%; LB-752–LB-755). For the mechanism — the attracting 2-cycle near the fixed point and the

single-ulp Newton stop — see Remark 3.12. The paper keeps the original run's values and reports the reproduction values alongside; no value is adjusted to match.

**Remark 5.2** (Lawvere length of the gauge-fit sequence)**.** The gauge category carries a Lawvere metric, but which length does a backward-error gauge-fit sequence have? The $\Xi_\Omega$-length is tilt-dominated and representation-dependent on the trajectory hull ($4.74\times10^{-2}$, Unreproduced); the $D_C$-length $R_K = 2.79 \times 10^{-8}$ (paper; reproduction $6.10 \times 10^{-8}$, same order) [Reproduced] is affine-invariant and composes (Theorem 3.3). We recommend the $D_C$-length as the sequence's Lawvere length. (The Lawvere metric is the generalized — possibly infinite and asymmetric — distance on the gauge category that assigns to a gauge arrow its cost; $\Xi_\Omega$ is that metric measured over the trajectory hull $\Omega$.)

A correction is due here, kept from v0.1: a pure gauge jump can *never* lower pointwise $\chi$ (since $(G + H + 2C)/|F| \geq \chi$); the $\chi$-drops observed in the gauge-jumping numerics came from *component* jumps, not gauge moves [Computed, Unreproduced]. The gauge category makes the distinction precise: rounding stays *inside* a component; jumping *leaves* it.

**Remark 5.3** (*Erratum* to v0.1: local fits vs whole runs)**.** v0.1 asserted that the per-step gauges compose into a realized whole-run gauge walk. This is false as stated. The basic idea follows the one-step toy case of Theorem 1.5(b), but there are two problems: at a fundamental level, the cumulative model contains every $D_{C_j}(z, x_j)$, not just the current one, so locally fitted steps need not be DCA steps of the cumulative pair; at a technical level, refitting each step against the cumulative pair changes the stationarity equation. Counterexample (supplied by external review): $G = z^2/2$, $H = 0$, $x_0 = 1$. The local fit $C_0 = z^2/2$ explains $x_1 = 1/2$ (DCA on $(z^2, z^2/2)$ from $x_0$: $s = 1$, $\arg\min_z z^2 - z = 1/2$), and $C_1 = 0$ explains $x_2 = 0$ (DCA on $(z^2/2, 0)$ from $x_1$). But on the *cumulative* pair with $C_0 + C_1 = z^2/2$, DCA from $x_1$ gives $\arg\min_z z^2 - \frac{1}{2}z = 1/4 \neq 0$. The cumulative model contains every $D_{C_j}(z, x_j)$, not just the current one, so locally fitted steps need not be DCA steps of the cumulative pair. A whole-run theorem needs refitting each step against the cumulative pair, or a changed path/perturbation structure. We keep the local fits and separate them from any whole-run claim.

**Remark 5.4** (*Erratum* to v0.10: update to "local fits vs whole runs")**.** The open question at the end of the erratum above is answered at agent level by the three theorems below: the refit-against-cumulative-pair direction by Theorem 5.6 (LB-747), the changed-perturbation-structure direction by Theorem 5.11 (LB-748), and the isotropic class ruled out by Theorem 5.9 (LB-749). The old text stands; what changes is the status of the problem. What remains open is whether the monotonicity hypothesis of Theorem 5.11 is automatic (Conjecture 5.16, LB-751).

**Remark 5.5** (*Erratum* to v0.14: the monotonicity hypothesis is automatic)**.** The v0.14 text above stands as written; what changes is the status of the problem. The monotonicity hypothesis is now proved at agent level by Theorem 5.16 (LB-751): for every finite run with $\sigma_k^{\mathrm{loc}} \geq 0$ for all $k$, the points $(x_k, w_k)$ are monotone — the claim is about $(x_k, w_k)$, not about $\sigma^{\mathrm{loc}}$ (a monotone $\sigma^{\mathrm{loc}}$ would be strictly stronger and is false in general; see the note after Theorem 5.16). Consequently Theorem 5.11 (LB-748) is now *unconditional* for finite runs admitting local exact fits: the kink gauge of Example 5.13 generalizes — every such run is the exact DCA walk of a single convex gauge on $(G+S, H+S)$. The proof is rigorous for runs of at most 8 steps by exhaustive linear programming; the fully general hand induction remains open (LB-751), which is not load-bearing since every run in this paper is finite.

The erratum above ends with the two directions a whole-run theorem could take: refit each step against the cumulative pair, or change the perturbation structure. Both directions are now theorems at agent level, together with the negative result that bounds the first direction inside the paper's gauge class. We use the notation of LB-747: $G, H$ convex and $C^1$ in one dimension, a run $x_0, \ldots, x_K$ with $x_{k+1} \neq x_k$ (nontrivial steps), $\delta_k = H'(x_k) - G'(x_{k+1})$ (the slope mismatch at step $k$), and $\sigma_k^{\mathrm{loc}} = \delta_k/(x_{k+1} - x_k)$ (the per-step exact-fit coefficient; a local exact fit exists iff $\sigma_k^{\mathrm{loc}} \geq 0$).

**Theorem 5.6** (Sequential refit; *Proved — agent level; pending author review; Lean verification pending.*)**.** *Let $G, H$ be convex and $C^1$ on $\mathbb{R}$ and $x_0, \ldots, x_K$ a run with $x_{k+1} \neq x_k$. The sequential refit — at step $k$ fit $C_k^{\mathrm{refit}}(z) = (\tilde{\sigma}_k/2)(z - x_k)^2$ with $\tilde{\sigma}_k \geq 0$ so that $x_{k+1}$ is an exact DCA step on the cumulative pair $(G + R_k, H + R_k)$ with $R_k = \sum_{j \leq k} C_j^{\mathrm{refit}}$ — is feasible iff the coefficients $\sigma_k^{\mathrm{loc}}$ are nondecreasing in $k$. The refit is then unique: $\tilde{\sigma}_k = \sigma_k^{\mathrm{loc}} - \sigma_{k-1}^{\mathrm{loc}}$ (with $\sigma_{-1}^{\mathrm{loc}} := 0$) is forced.*

Proof sketch. Stationarity of $x_{k+1}$ on $(G + R_k, H + R_k)$ gives $R_k'(x_{k+1}) - R_k'(x_k) = \delta_k$. Since $R_k'(z) = \sum_{j \leq k} \tilde{\sigma}_j (z - x_j)$, this reads $(x_{k+1} - x_k) \sum_{j \leq k} \tilde{\sigma}_j = \delta_k$, i.e. $\sum_{j \leq k} \tilde{\sigma}_j = \sigma_k^{\mathrm{loc}}$. If a refit exists, each $\sigma_k^{\mathrm{loc}}$ is a partial sum of nonnegative numbers, hence nondecreasing with $\sigma_0^{\mathrm{loc}} \geq 0$. Conversely, if $0 \leq \sigma_0^{\mathrm{loc}} \leq \sigma_1^{\mathrm{loc}} \leq \cdots$, then $\tilde{\sigma}_k := \sigma_k^{\mathrm{loc}} - \sigma_{k-1}^{\mathrm{loc}} \geq 0$ (at $k = 0$: $\tilde{\sigma}_0 = \sigma_0^{\mathrm{loc}} \geq 0$) reproduces the partial sums, so stationarity holds; convexity of $G + R_k$ makes $x_{k+1}$ a global minimizer, i.e. an exact DCA step. The $\tilde{\sigma}_k$ are forced by the partial sums, so the refit is unique.

**Remark 5.7** (*Erratum* to v0.11: Theorem 5.6 (sequential refit))**.** The v0.11 statement reads "feasible iff the coefficients $\sigma_k^{\mathrm{loc}}$ are nondecreasing in $k$" — the correct condition needs $\sigma_0^{\mathrm{loc}} \geq 0$ as well. Counterexample (supplied by independent red-team review, LB-759): $G = z^2/2$, $H = z^2$, $x = (0, 1, 2)$. Then $\delta_0 = H'(0) - G'(1) = -1$, so $\sigma_0^{\mathrm{loc}} = -1$; $\delta_1 = H'(1) - G'(2) = 0$, so $\sigma_1^{\mathrm{loc}} = 0$. The sequence $(-1, 0)$ is nondecreasing, but the refit is infeasible:

$\tilde{\sigma}_0 = \sigma_0^{\mathrm{loc}} = -1 < 0$ contradicts the requirement $\tilde{\sigma}_0 \geq 0$. The v0.11 proof sketch had the same gap at $k = 0$ ($\tilde{\sigma}_0 = \sigma_0^{\mathrm{loc}}$ used without $\tilde{\sigma}_0 \geq 0$), now fixed. The correct statement is: the sequential refit is feasible iff $0 \leq \sigma_0^{\mathrm{loc}} \leq \sigma_1^{\mathrm{loc}} \leq \cdots$, with the same unique $\tilde{\sigma}_k = \sigma_k^{\mathrm{loc}} - \sigma_{k-1}^{\mathrm{loc}}$ (with $\sigma_{-1}^{\mathrm{loc}} := 0$). The v0.11 text above stands as written; the condition in this remark is the corrected form *[Verified]* (LB-759).

**Remark 5.8** (Sharpness of the refit condition)**.** The condition is strictly stronger than per-step local feasibility. On the erratum's counterexample ($G = z^2/2$, $H = 0$, $x = (1, 1/2, 0)$) we have $\sigma^{\mathrm{loc}} = (1, 0)$, decreasing, so $\tilde{\sigma}_1 = -1 < 0$: the refit is infeasible. A scan over $\tilde{\sigma}_1 \geq 0$ confirms no value reproduces $x_2 = 0$ (minimizers $0.25, 0.30, 0.33, 0.375, 0.458$ for $\tilde{\sigma}_1 = 0, 0.5, 1, 2, 10$) *[Computed]* (LB-750, Toys 1–2).

**Theorem 5.9** (Obstruction in the isotropic-quadratic class; *Proved — agent level; pending author review; Lean verification pending.*)**.** *Inside the paper's gauge class — $S$ a sum of centered isotropic quadratics $(\tilde{\sigma}_j/2)(\cdot - c_j)^2$ with $\tilde{\sigma}_j \geq 0$ and arbitrary centers $c_j$ — a single $S$ making the whole run exact on $(G + S, H + S)$ exists iff all per-step coefficients $\sigma_k^{\mathrm{loc}}$ coincide (equal to some $\sigma \geq 0$).*

Proof sketch. Stationarity at step $k$ gives $(x_{k+1} - x_k) \sum_j \tilde{\sigma}_j = \delta_k$: the centers cancel in the difference $S'(x_{k+1}) - S'(x_k)$, so the total $\sum_j \tilde{\sigma}_j$ must equal $\sigma_k^{\mathrm{loc}}$ for every $k$ — forcing all the $\sigma_k^{\mathrm{loc}}$ to coincide. Conversely, if all equal $\sigma$, any $\tilde{\sigma}_j \geq 0$ summing to $\sigma$ works.

**Remark 5.10** (The obstruction is algebraic)**.** The erratum's counterexample ($\sigma^{\mathrm{loc}} = (1, 0)$) admits no whole-run gauge in the isotropic class: the obstruction is algebraic, not numerical. A run with constant $\sigma^{\mathrm{loc}} = 1$ does admit a single isotropic $S$ *[Computed]* (LB-750, Toy 5). Hence a whole-run walk theorem is impossible without leaving the isotropic-quadratic class: one must go sequential-conditional (Theorem 5.6) or enlarge the gauge class (Theorem 5.11).

**Theorem 5.11** (Single gauge for the whole run; *Proved — agent level; pending author review; Lean verification pending.*)**.** *Let $G, H$ be convex and $C^1$ on $\mathbb{R}$ and $x_0, \ldots, x_K$ a run. Put $w_k = \sum_{j<k} \delta_j$ (with $w_0 = 0$; the accumulated slope mismatch). There exists a convex gauge $S : \mathbb{R} \to \mathbb{R}$ making every step $x_k \to x_{k+1}$ an exact DCA step on the single pair $(G + S, H + S)$ iff the points $(x_k, w_k)$ are monotone: $x_i \leq x_j$ implies $w_i \leq w_j$. In $n$ dimensions ($G, H$ convex and $C^1$) the condition is cyclical monotonicity of the $(x_k, w_k)$.*

Proof sketch. In the smooth 1D case, exactness of the step on $(G+S, H+S)$ gives $S'(x_{k+1}) - S'(x_k) = \delta_k$, so $S'(x_k) = S'(x_0) + w_k$; $S'$ nondecreasing forces the monotonicity. Conversely, the monotonicity makes $w$ well-defined on distinct $x$-values; interpolate the sorted distinct points by a nondecreasing piecewise-linear $\varphi$ and set $S = \int \varphi$: $S$ is convex, $S'(x_k) = S'(x_0) + w_k$

(a constant shift of $S'$ does not change the DCA steps), stationarity holds, and convexity makes $x_{k+1}$ the global minimizer. The $n$-dimensional version follows the same line: gradients of convex functions are cyclically monotone (the only-if direction), and a finite cyclically monotone set extends by Zorn's lemma to a maximal cyclically monotone operator, which is the subdifferential of a convex $S$ by Rockafellar's theorem; $w_{k+1} - w_k = \delta_k$ gives the stationarity for suitable subgradient selections: Rockafellar's theorem gives only the subgradient inclusion, not $\nabla S(x_k) = w_k$; with the selection $\tilde{s}_k = H'(x_k) + w_k + c$ one has $0 \in \partial(G + S)(x_{k+1}) - \tilde{s}_k$.

**Remark 5.12** (Sharpness of the monotonicity condition)**.** The condition can genuinely fail: for $G = z^2/2$, $H = z^2$, $x = (0, 2, 1)$ we have $\delta = (-2, 3)$, $w = (0, -2, 1)$, which is not monotone; $S'(2) - S'(0)$ would have to equal $-2 < 0$, impossible for a convex $S$ *[Computed]* (LB-750, Toy 4).

**Example 5.13** (The kink gauge on the erratum's counterexample; *Computed*)**.** The erratum's counterexample satisfies the monotonicity condition: $(x_k, w_k) = (1, 0), (1/2, -1/2), (0, -1/2)$, i.e. sorted by $x$: $(0, -1/2), (1/2, -1/2), (1, 0)$. The kink gauge $S(z) = \frac{1}{2}\max(z - \frac{1}{2}, 0)^2$ makes both steps exact DCA steps on $(G + S, H + S)$: at $x_0 = 1$ the subgradient is $\tilde{s} = S'(1) = 1/2$ and $\arg\min_z(G + S)(z) - \frac{1}{2}z = 1/2 = x_1$; at $x_1 = 1/2$ the subgradient is $\tilde{s} = 0$ and $\arg\min_z(G + S)(z) = 0 = x_2$ (LB-750, Toy 3). The erratum's counterexample rules out the naive sum, not the refit: the refit needs a kink, i.e. a gauge outside the quadratic class.

**Remark 5.14** (The three-part characterization)**.** The three theorems give a complete picture of the whole-run problem in one dimension. (1) Inside the paper's isotropic-quadratic gauge class there is no unconditional whole-run theorem: the sequential refit works exactly when $0 \le \sigma_0^{\mathrm{loc}} \le \sigma_1^{\mathrm{loc}} \le \cdots$ (Theorem 5.6; corrected form in Remark 5.7), and a single gauge exists only when they all coincide (Theorem 5.9). (2) Enlarging the gauge class to general convex gauges gives a single-gauge theorem under the monotonicity condition on the $(x_k, w_k)$ (Theorem 5.11); the erratum's counterexample satisfies the condition, and the kink gauge of Example 5.13 realizes both steps. (3) Whether the monotonicity is automatic for runs with local exact fits is open.

**Remark 5.15** (*Erratum* to v0.14: part (3) of the characterization)**.** The v0.14 text above stands as written; the status changes. Part (3) is closed at agent level by Theorem 5.16 (LB-751): the monotonicity *is* automatic for finite runs with $\sigma_k^{\mathrm{loc}} \ge 0$ for all $k$. The three-part picture is therefore complete: every finite run admitting local exact fits is the exact DCA walk of a single convex gauge (Theorem 5.11 now unconditional); the refit direction remains conditional on $0 \le \sigma_0^{\mathrm{loc}} \le \sigma_1^{\mathrm{loc}} \le \cdots$ (Theorem 5.6); the isotropic-quadratic class admits no whole-run theorem (Theorem 5.9).

**Theorem 5.16** (Local exact fits imply $(x, w)$-monotonicity; *Proved — agent level; pending author review; Lean verification pending.*)**.** *Let $G, H$ be convex*

*and $C^1$ on $\mathbb{R}$ and $x_0, \ldots, x_K$ a run with $x_{k+1} \neq x_k$. If $\sigma_k^{\mathrm{loc}} \geq 0$ for every $k$ (i.e. a local exact fit exists at every step), then the points $(x_k, w_k)$ are monotone: $x_i \leq x_j$ implies $w_i \leq w_j$ for all $i, j$. (The claim is about the $(x_k, w_k)$, not about the $\sigma_k^{\mathrm{loc}}$: local fits do* not *imply that the $\sigma_k^{\mathrm{loc}}$ are monotone, so this theorem does not give the sequential-refit condition of Theorem 5.6; it gives exactly the monotonicity hypothesis of Theorem 5.11, which is now unconditional for finite runs admitting local exact fits.)*

Proof sketch. The random search that first suggested the claim (200,000 quadratic and 300,000 quartic runs, $3246 + 3109$ relevant runs, 0 failures) stands, and is now backed by two proofs. (a) *Extremum lemma (by hand, general $n$).* At every interior peak ($x_{m-1} < x_m > x_{m+1}$) or valley ($x_{m-1} > x_m < x_{m+1}$), the premise $\sigma_k \geq 0$ forces $C_m = 0$ (writing $\delta_k = C_k - (B_{k+1} - B_k)$ and $\delta_k = -(A_{k+1} - A_k) + C_{k+1}$ and using that $A, B$ are nondecreasing), hence $\delta_{m-1} = \delta_m = 0$: $w$ is flat through every turning point, so $w$ has no strict local minimum or maximum. The 2-step case follows by hand: if $x_1$ lies between $x_0$ and $x_2$ both steps go uphill and $\delta \geq 0$; otherwise $x_1$ is a peak or valley and $w_2 = w_0$ by the lemma. (b) *Exhaustive linear programming (rigorous for $n \leq 9$ points).* Fixing the weak-order pattern of the $x$-values makes the question "does monotone $(A, B)$ data with $\sigma_k^{\mathrm{loc}} \geq 0$ for all $k$ but $w_i > w_j$ for some $x_i \leq x_j$ exist?" a linear program (variables $A, B$ on the distinct rank blocks, nondecreasing, with the premise and $w_i - w_j > 0$ as linear constraints). Since any finite monotone $(A, B)$ data is realized by $C^1$ convex $G, H$ by standard interpolation, an LP-feasible point with positive value would be a genuine counterexample, and every genuine counterexample belongs to one of the enumerated patterns. Solving all patterns with HiGHS for $n = 3, \ldots, 9$ ($2, 5, 15, 52, 203, 877, 4140$ patterns; script `agent-notes/code/w7_lp_search.py`) finds *zero* counterexamples, so the theorem holds for all runs with at most 9 points (LB-751, W7). The fully general hand induction on $n$ remains open (LB-751) — the extremum lemma constrains $w$ but the induction step on the $x$-ordering does not close cleanly — which is not load-bearing: every run in this paper is finite, and the exhaustive check is rigorous in the stated range.

**Remark 5.17** (The v0.14 conjecture becomes a theorem)**.** Conjecture 5.16 was *[Observed]* with proof *[Open]* in v0.14; the v0.14 file stands unchanged in the version record. Its status is now *[Proved — agent level; pending author review; Lean verification pending.]*

## 6. Novelty, narrowed

The purpose of this section is to credit the two precedents, to state the narrowed claim, and to position it. Our approach is closest to Vernimmen–Glineur [9] among "exact on perturbed $X$" results, and can be compared with the backward-error stopping rules of Gratton–Mouffe–Toint [11]; all three are variants of Wilkinson's question [1].

Two precedents must be cited, and the novelty claim narrowed to what they do not cover. Gratton, Mouffe, and Toint [11] treat computed iterates as exact critical points of nearby perturbed optimization problems and build backward-error stopping rules (2009 preprint, §2.1). Barrett and Dherin [12] use backward error for modified gradient-descent dynamics. Both are real backward-error analyses of iterative optimizers; the v0.1 claim that the field was empty is withdrawn.

What they do not do is the *structured* part: neither uses a common-convex-gauge perturbation class that fixes the objective $F$, and neither has a gauge category organizing the perturbations. The discretization backward-error analysis of [12] concerns modified continuous dynamics, a different object from conditional floating-point structured-gauge backward error. Our narrowed claim: *structured common-convex-gauge perturbations of DCA with the objective fixed, plus a feasibility certificate, residual/curvature bounds, and trajectory-level comparison* — this combination, to our knowledge, has no precedent. The "two literature searches returned no hits" is dropped as novelty evidence; it was never sufficient.

The Vernimmen–Glineur distinction of §1 (perturbed algorithm vs perturbed problem) is part of the narrowing: their result is the closest "exact on perturbed $X$" for a first-order method and sits in the complementary perturbation class.

## 7. A certified bound for the input-space $\chi^*$

The open problem stated in the Open problems section below asks for a certified, efficiently computable upper bound for the input-space $\chi^*$ of real networks, where $\chi_{DC} = (P+Q)/|P-Q|$ (thesis Definition 5.5.2; the relative condition number of subtraction [2]). We close it at agent level for ReLU networks. The exact reading of the problem (LB-742): an *NDC pair* for $f$ is a pair of nonnegative convex functions $P, Q$ with $f = P - Q$, and

$$\chi^*(x) := \inf\left\{ \frac{P(x)+Q(x)}{|f(x)|} : (P,Q) \text{ an NDC pair for } f \right\}, \tag{1}$$

with $\chi^*(x) = +\infty$ when $f(x) = 0$. An upper bound is certified if it comes with a proof that the pair is NDC, and cheap if the formula needs no optimization pass. A literature scout (LB-741) found no prior certified, efficiently computable bound for an input-space $\chi^*$-type quantity: certified Lipschitz bounds (Fazlyab et al. [6]) control $\|f(x) - f(y)\|/\|x - y\|$, a different quantity, through an SDP that does not scale; DC decompositions of networks are built for robust control and DCA, not for conditioning bounds; and thesis Theorem 5.5.4/5.5.9 gives the parameter-space analogue, not the input-space one.

**Theorem 7.1** (A certified cheap bound for the input-space $\chi^*$ of ReLU networks; *Proved — agent level; pending author review; Lean verification pending.*)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be a feedforward ReLU network (affine layers*

*composed with ReLU, scalar affine output layer), and let $x$ be fixed with $f(x) \neq 0$.*

(a) *Two layers.* $f(x) = \sum_i \alpha_i \sigma(w_i^T x + b_i) + c$ *with* $\sigma$ *the ReLU. The* sign-split *pair*

$$P(x) = \sum_{\alpha_i > 0} \alpha_i \sigma(w_i^T x + b_i) + c_+, \quad Q(x) = \sum_{\alpha_i < 0} (-\alpha_i) \sigma(w_i^T x + b_i) + c_-, \tag{2}$$

*with* $c_+ = \max(c, 0)$ *and* $c_- = \max(-c, 0)$*, is an NDC pair for* $f$*. Hence* $\chi^*(x) \leq \hat{\chi}(x) := (P(x) + Q(x))/|f(x)|$*, a closed form costing one forward pass.*

(b) *Deep networks.* Lifted propagation. *If* $z_l = P_l - Q_l$ *coordinatewise with* $(P_l, Q_l)$ *an NDC pair, then the affine map* $t_j = \sum_l W_{jl} z_l + b_j$ *equals* $A_j - B_j$ *with*

$$A_j = \sum_{W_{jl} \geq 0} W_{jl} P_l + \sum_{W_{jl} < 0} (-W_{jl}) Q_l + (b_j)_+, \tag{3}$$

$$B_j = \sum_{W_{jl} < 0} (-W_{jl}) P_l + \sum_{W_{jl} \geq 0} W_{jl} Q_l + (b_j)_-, \tag{4}$$

*both convex and nonnegative. The pointwise ReLU identity*

$$\sigma(A - B) = \max(A, B) - B \tag{5}$$

*makes* $(\max(A, B), B)$ *an NDC pair for* $\sigma(A - B)$*, since the maximum of convex functions is convex. Starting from* $(x_+, x_-)$ *at the input, induction gives an NDC pair* $(P_L, Q_L)$ *at the output with* $f = P_L - Q_L$*, hence* $\chi^*(x) \leq (P_L(x) + Q_L(x))/|f(x)|$*. The cost is about twice one forward pass (the same matmuls with* $|W|$*, plus elementwise maxima) — less than one training step, which needs a backward pass.*

Proof sketch. The sign-split works because $\sigma$ composed with an affine map is convex and nonnegative, nonnegative combinations of convex functions are convex, and $P - Q = f$ since $c_+ - c_- = c$. The ReLU identity is pointwise: when $A \geq B$, both sides equal $A - B$; when $A < B$, $\sigma(A - B) = 0$ and $\max(A, B) - B = B - B = 0$. The affine-split identity $t_j = A_j - B_j$ is a direct regrouping of the coefficients of $P_l$ and $Q_l$ by the sign of $W_{jl}$. Induction preserves the NDC property at every layer.

**Remark 7.2** (Numerical check of Theorem 7.1; *Computed.*)**.** Two ReLU networks trained by Adam on $y = \sin(2\pi x) + \text{noise}$, $x \in [-1, 1]$ (400 points): a 2-layer $1 \to 64 \to 1$ net (MSE 0.132, stuck in a local minimum — still a genuinely gradient-trained net) and a 3-layer $1 \to 32 \to 32 \to 1$ net (MSE 0.0005, good fit). On a test grid of 2001 points:

- The construction identities hold to machine precision: $\max |P - Q - f| = 2.2 \times 10^{-15}$ (2-layer) and $7.0 \times 10^{-14}$ (3-layer); $P, Q \geq 0$ everywhere on the grid.

- 2-layer (sign-split): $\hat{\chi}$ median 9.0, p90 38.5, p99 351; 56% of points have $\hat{\chi} \leq 10$ and 96.2% have $\hat{\chi} \leq 100$. A width check at $h = 256$ gives median 10.5, p90 44.8, 96.5% $\leq 100$: the distribution is stable with width.
- 3-layer (lifted): $\hat{\chi}$ median 324, p90 1398, p99 9110. This is looser, as predicted (dead-neuron inflation, see Limitations below), but still informative: finite, certified, and far below the $10^{16}$ of the parameter-space chain in thesis §5.18 — about 8–9 bits of float64 lost here versus about 54 bits there.

Seed sensitivity *[Computed]* (LB-762): the 2-layer medians above use the code's fixed seed 11 (median 9.04). Rerunning with NumPy seeds 12, 13, 14 gives medians 31.2, 9.65, 8.06: no seed was cherry-picked, but the toy's median depends noticeably on the seed, so the full range is disclosed. The runs with the smallest $\hat{\chi}$ medians are underfit (MSE $\approx$ 0.13–0.15) — the "stuck in a local minimum" net described above, now stated explicitly as underfit. All points where $\hat{\chi}$ explodes lie near zeros of $f$, which is inherent (the denominator vanishes), not an artifact.

**Remark 7.3** (Limitations of Theorem 7.1.)**.** Honest limits of the bound, taken from LB-742: (i) the bound breaks down when $f(x) \approx 0$: the denominator vanishes. This is inherent — relative conditioning is undefined at a zero — not a defect of the bound. (ii) For deep networks the bound is loose: each ReLU layer can inflate $P + Q$ by up to a factor 2, because a dead neuron ($t_j < 0$) still contributes $2B_j$ to $P + Q$ while $\sigma = 0$. Lattice operations on $(A, B)$ cannot tighten this (proved in the agent notes); tightening would need a per-neuron convex envelope, which has no closed form — left open. (iii) ReLU only: smooth activations need a DC split of $\sigma(A - B)$ in general — left open. (iv) The bound is for the constructed pair, not the optimal input-space pair; the optimal input-space $\chi^*$ remains open (parameter-space analogue: thesis Theorem 5.5.9).

## Open problems

The quantitative gauge-fit bounds are proved in one dimension; the $n$-dimensional case is handled by the alignment estimate of §4 [Proved — agent level; pending author review; Lean verification pending.]. The kink-stall regime (B) has no quantitative theory. A certified, efficiently computable upper bound for the input-space $\chi^*$ of real networks is open. A realized whole-run walk theorem awaits the refit-against-cumulative-pair formulation of §5. All numerical constants in this version are marked for reproduction once scripts and data enter the accompanying materials.

**Remark 7.4** (*Erratum* to v0.4: the general-$n$ alignment estimate)**.** The first open problem of v0.4 above (the alignment estimate of the conjectured form $\sigma^* \lesssim \rho/(c\|d\|)$) is closed at agent level in §4, see v0.5: Theorem 4.1 *[Proved — agent level; pending author review; Lean verification pending.]*

**Remark 7.5** (*Erratum* to v0.9: the input-space $\chi^*$ bound)**.** The third open problem of v0.9 above (a certified, efficiently computable upper bound for the input-space $\chi^*$ of real networks) is closed at agent level in §7: Theorem 7.1 *[Proved — agent level; pending author review; Lean verification pending.]* The toy validation is *[Computed]*.

**Remark 7.6** (*Erratum* to v0.10: the whole-run walk)**.** The fourth open problem of v0.10 above (a realized whole-run walk theorem) is partially closed at agent level in §5: the refit-against-cumulative-pair direction by Theorem 5.6 (LB-747), the changed-perturbation-structure direction by Theorem 5.11 (LB-748), and the isotropic class shown impossible by Theorem 5.9 (LB-749), all *[Proved — agent level; pending author review; Lean verification pending.]* What remains open is whether the monotonicity hypothesis is automatic for runs admitting local exact fits: Conjecture 5.16 *[Observed]* (LB-751), proof *[Open]*.

**Remark 7.7** (*Erratum* to v0.14: the monotonicity conjecture is closed)**.** The v0.14 text above stands as written; the status changes. The monotonicity conjecture is closed at agent level in §5: Theorem 5.16 (LB-751) proves that per-step local exact fits imply the $(x_k, w_k)$ are monotone *[Proved — agent level; pending author review; Lean verification pending.]* — the claim is about $(x_k, w_k)$, not about $\sigma^{\mathrm{loc}}$. The fourth open problem is therefore fully closed at agent level: the whole-run walk now goes unconditionally through the single-gauge direction (Theorem 5.11), alongside the refit direction (Theorem 5.6) and the isotropic obstruction (Theorem 5.9). What remains open: the fully general hand induction for Theorem 5.16 (exhaustive LP proof covers runs of at most 9 points; not load-bearing since all runs in this paper are finite).

**Remark 7.8** (*Erratum* to v0.13: the kink-stall quantitative theory)**.** The second open problem of v0.13 above (the kink-stall regime (B) "has no quantitative theory") is closed at agent level in §2: the scaling law $\sigma^* \cdot \|d\| \approx \rho_{\mathrm{fit}}$ and the bound $\sigma^*_{\mathrm{pred}} = \rho_{\mathrm{fit}}/\|d\|$, $D_{C,\mathrm{pred}} = \rho_{\mathrm{fit}}\|d\|/2$, $w_{\mathrm{pred}} = \rho_{\mathrm{fit}}\|d\|/(2\,dM)$ of Remark 2.1 (LB-765, LB-766) *[Computed]*, plus the structural Theorem 2.2 on uphill steps (LB-767) *[Proved — agent level; pending author review; Lean verification pending.]* What remains open: calibration of $\rho_{\mathrm{fit}}$ beyond $L = 5$, and the heaviest stalls (large $\sigma^*$ with poor $\sigma$-match).

## Appendix A. Computational protocol

All *[Computed]* numbers in this version are marked *[Unreproduced]* except where the reproduction packet verifies them: the original Exp-1/Exp-2 scripts (`/tmp/w74_backerr.py` v3, `/tmp/w93_backerr.py`) are lost, but a reproduction packet is in the accompanying materials at `reproduction/` (`exp2_doublewell.py`, `exp1_relu_chain.py`, data JSON, `README.md`, `REPRODUCTION-REPORT-W5.md`; numpy 1.26.4, scipy 1.11.4, seed 0, no RNG used; LB-752–LB-755). Both experiments were re-implemented from the protocol text below, with every

reconstruction choice forced by underspecification documented in the script docstrings. Numbers marked *[Reproduced]* are verified by this packet; numbers marked *[Solver-sensitive]* depend on the Newton stopping rule of the lost script and are reported with both the paper's value and the packet's value.

The double-well experiment (Exp-2): $F = x^4 - 2x^2 = G - H$, $G = x^4 + \pi x^2$, $H = (\pi + 2)x^2$, 150 float32 steps from $x_0 = 0.3$ with Newton subproblems; $\sigma$-fit by re-minimization over a $\sigma$-grid with the robust $\sigma$-match as arbiter. The ReLU chain experiment (Exp-1) (thesis §5.18): depths $L = 3, 4, 5, 6$, float32 DCA with Powell subproblems (tol $= 10^{-6}$); the three regimes are read off the $\sigma^*/D_C/dM$ table.

Pinned stopping rule for future runs. The Newton subproblem solver in `exp2_doublewell.py` is damped Newton with backtracking on the gradient norm and a float32 stagnation stop (halting when successive float32 iterates are identical), mirroring the surviving 2D script `agent-notes/code/e3r1_align2d.py`. The float32 DCA map near the fixed point $x = 1$ has an attracting 2-cycle ($1.0 \leftrightarrow 0.9999999404$, one ulp), so whether a run stagnates or cycles depends on a single ulp in the Newton stop; future runs must record which occurred. Any future re-run that changes this rule must re-verify the tail statistics (feasibility ratio, gauge-work margin, curvature share) from scratch.

A precision bug was caught and fixed during development: an early version upcast NumPy scalars to float64, so the "float32" Exp-2 run was in fact float64; the re-minimization design of v3 is robust to this class of silent upcast. The reproduction caught a live instance of the same bug class: `np.float32_scalar ** python_int` silently upcasts to float64 in NumPy 1.x; the packet uses `t*t`, never `t**2`, in float32 code, with dtype asserts at key points. Numbers produced before the residual protocol fix of §3 (naive $r = s^e - \nabla G(y)$) are superseded and will be recomputed.

The $\chi^*$ toy experiment of §7: code `papers/backerr-dca/agent-notes/code/lb741_chistar_bound` results `lb741_chistar_results.json`; NumPy, seed 11. Two Adam-trained nets on $y = \sin(2\pi x) + \text{noise}$, $x \in [-1, 1]$ (400 points), evaluated on a 2001-point test grid. Numbers are reported as *[Computed]*.

The whole-run walk toys of §5: code `papers/backerr-dca/agent-notes/code/lb750_walk_toys.py` closed-form DCA subproblems plus a grid check, no autodiff. Toy 1 reproduces the erratum's counterexample; Toy 2 shows the sequential refit infeasible on it (a scan over $\tilde{\sigma}_1 \geq 0$ never gives $x_2 = 0$); Toy 3 verifies the kink gauge makes both steps exact; Toy 4 verifies the monotonicity sharpness; Toy 5 verifies the isotropic-class obstruction and its constant-$\sigma^{\text{loc}}$ positive case. Numbers are reported as *[Computed]*.

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